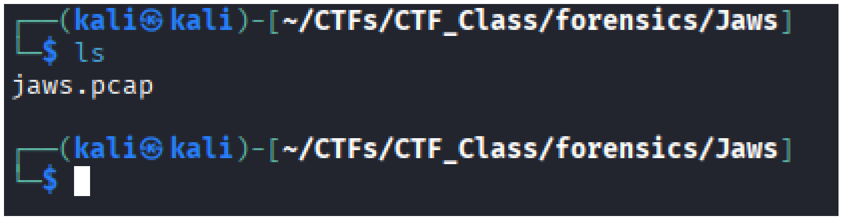
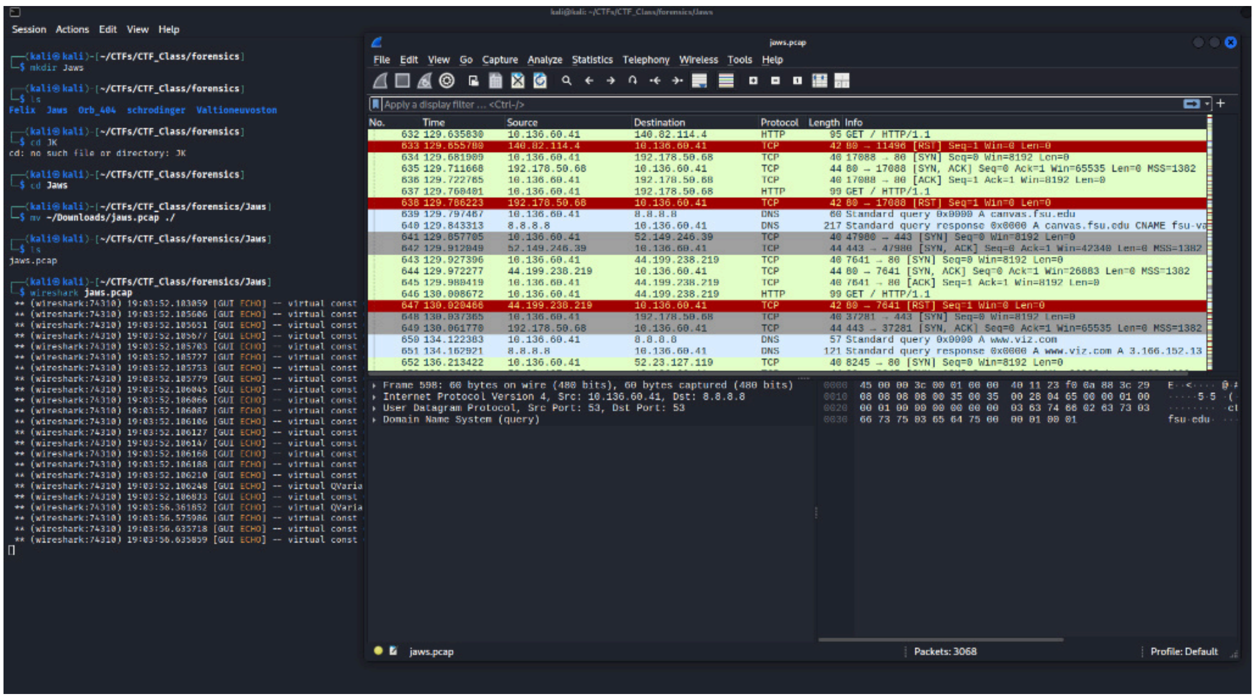
Problem ID	Captured Flags	Steps
jaws	fsuCTF{Y0ur3_G0nn4_N33d_A_Bi6g3r_8o47}	1. Open in wireshark 2. Filter for http requests 3. Find malicious GET request 4. Inspect the AP traffic 5. Find the flag using tshark
whats_the_title_again?	fsuCTF{1_d0n7_w4n7_t0_pl4y_4nym0r3}	1. Look at the session data 2. Notice wmplayer.exe 3. Grep for the process and grep it with .mp3 for a sound file 4. Search the raw mtf and original file names surrounding the Sleep Away.mp3 file in the data
Xperience	fsuCTF{p1ca55o_w0u1dv3_l0v3d_m5pa1n7}	1. Open the 7zip and notice it is a hard drive 2. Navigate the hard drive using fsl 3. Find the partition table and use that to find the daniel folder in the hard drive 4. Open the “My documents” folder and find the hint in the readme 5. Grep for a file called “masterpiece” 6. Extract the file using icat open it for the flag

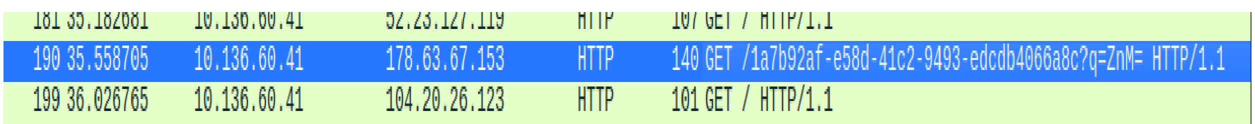
Jaws



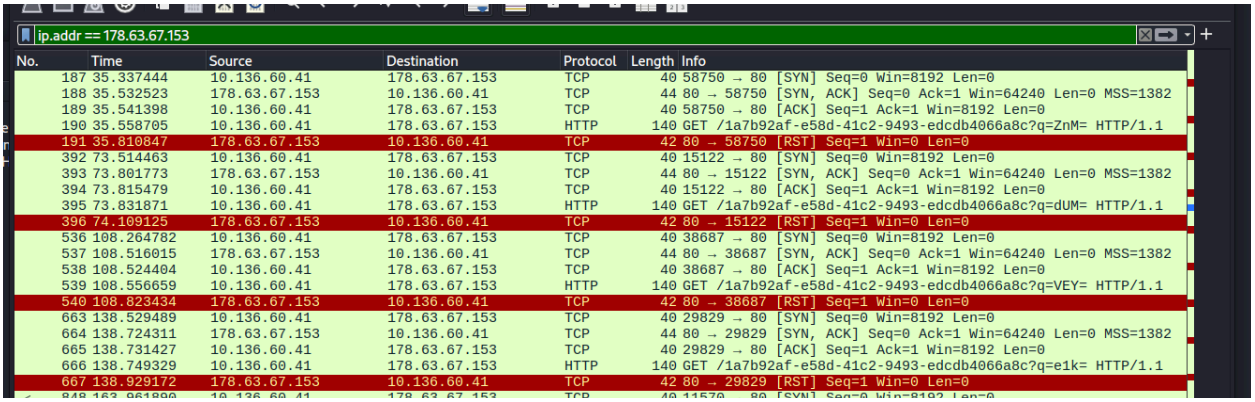
I first am given/download the following pcap file. I know to open this file in wireshark so I do:

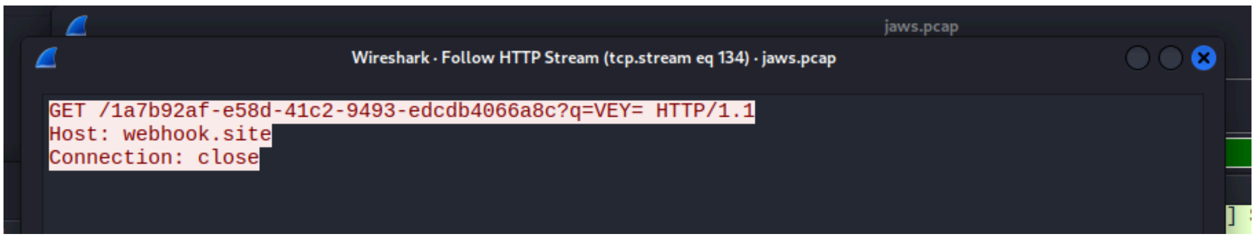


And I am met with loads of packet data so I first narrow it down to the http get and post requests:

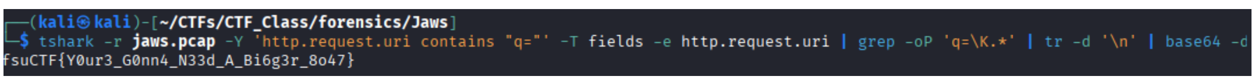


And this weird one kept popping up over and over so I check everything coming to and from that IP:



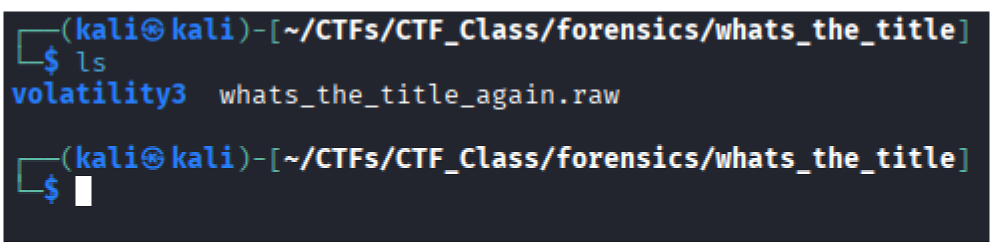


And it turns out it is a webhook that contains parts of the flag with the option q=___. So I plug this information into gemini to have me easily extract the flag from the pcap file which is:

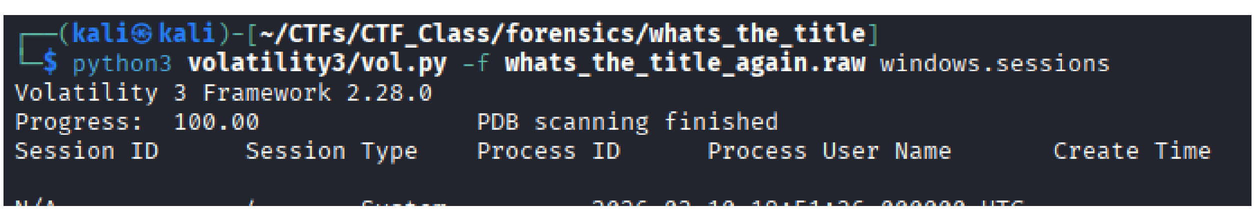


And I get the flag: fsuCTF{Y0ur3_G0nn4_N33d_A_Bi6g3r_8o47}

What's the title again?



I am given a memory dump. I downloaded volatility3 to help me navigate this memory. First what I do is look at the processes ran:



And found:

```
1       -       1224    rundll32.exe    WINDOWS7/vboxuser       2026-02-10 16:51:56.000000 UTC
1       -       1424    rundll32.exe    WINDOWS7/vboxuser       2026-02-10 16:51:56.000000 UTC
1       -       564     regsvr32.exe    -       2026-02-10 16:51:57.000000 UTC
1       Console 2004    wmplayer.exe    WINDOWS7/vboxuser       2026-02-10 16:54:39.000000 UTC
1       Console 2116    MineSweeper.ex  WINDOWS7/vboxuser       2026-02-10 16:54:48.000000 UTC
1       Console 2276    notepad.exe     WINDOWS7/vboxuser       2026-02-10 16:54:56.000000 UTC

┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/whats_the_title]
└─$ 
```

So wmplayer.exe was running which is an audio player. Lets see if we can find this process:

```
┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/whats_the_title]
└─$ strings -el whats_the_title_again.raw | grep -i "wmplayer.exe" | grep -i ".mp3"
"C:\Program Files\Windows Media Player\wmplayer.exe" /Relaunch /Play "C:\Users\Public\Music\Sample Music\Sleep Away.mp3"

┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/whats_the_title]
└─$ 
```

We found the process /Play which ran Sleep Away.mp3. But this is not the flag. I was told as a hint that you can change the name of the song before you run it so I decided to look in the data SURROUNDING Sleep Away.mp3 in the dump:

```
┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/whats_the_title]
└─$ strings -el whats_the_title_again.raw | grep -i -B 5 -A 5 "Sleep Away"
```

I use this command because -el flag is the magic key that exposes the raw MFT records and original filenames. I also use -A and -B to give me the data around Sleep Away 5 before and 5 after. And it ended up being an insane amount of data. I was also told I would "know it when I see it" so I decided to pipe it to a grep command filter for "_" because a flag ALWAYS has "_." And I find:

```
┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/whats_the_title]
└─$ strings -el whats_the_title_again.raw | grep -i -B 5 -A 5 "Sleep Away" | grep "_"
AMGa_id=R  1447875;AMGp_id=P   224157;AMGt_id=T 15061085
dows\winsxs\FileMaps\$$_system32_tasks_microsoft_b7abd682baafefc2.cdf-ms
1_d0n7_w4n7_t0_pl4y_4nym0r3
RunDLL32.EXE Shell32.DLL,ShellExec_RunDLL ?0x%X?%s
C:\Windows\WinSxS\x86_microsoft.windows.common-controls_6595b64144ccf1df_6.0.7601.17514_none_41e6975e2bd6f2b2\
1_d0n7_w4n7_t0_pl4y_4nym0r3
1_d0n7_w4n7_t0_pl4y_4nym0r3
 !"#$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxy
1_d0n7_w4n7_t0_pl4y_4nym0r3
1_d0n7_w4n7_t0_pl4y_4nym0r3
```

Flag: fsuCTF{1_d0n7_w4n7_t0_pl4y_4nym0r3}

Xperience

```
┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/Xperience]
└─$ ls
xp_smol.7z

┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/Xperience]
└─$ 7z x xp_smol.7z

7-Zip 25.01 (x64) : Copyright (c) 1999-2025 Igor Pavlov : 2025-08-03
 64-bit locale=en_US.UTF-8 Threads:16 OPEN_MAX:1024, ASM

Scanning the drive for archives:
1 file, 1951762005 bytes (1862 MiB)

Extracting archive: xp_smol.7z
--
Path = xp_smol.7z
Type = 7z
Physical Size = 1951762005
Headers Size = 130
Method = LZMA2:26
Solid = -
Blocks = 1

Everything is Ok

Size:       2903177216
Compressed: 1951762005

┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/Xperience]
└─$ ls
xp_smol.7z  xp_smol.vhd

┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/Xperience]
└─$ 
```

First we are given a 7zip file which I instinctively open using the 7z x command. So now I am given a .vhd file which I COULD mount but instead I found the tools “fls” and “icat” which searches the drive without mounting it making it much easier and efficient. First I want to see everything that is in this drive and where to access it so I run:

```
┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/Xperience]
└─$ fls xp_smol.vhd
Possible encryption detected (High entropy (7.72))
```

This means I need an offset to find the actual file system so I use mmls to give the partition table which is:

```
┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/Xperience]
└─$ mmls xp_smol.vhd
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

      Slot      Start        End          Length       Description
000:  Meta      0000000000   0000000000   0000000001   Primary Table (#0)
001:  -------   0000000000   0000000062   0000000063   Unallocated
002:  000:000   0000000063   0008369864   0008369802   NTFS / exFAT (0x07)
003:  -------   0008369865   0008388607   0000018743   Unallocated

┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/Xperience]
└─$ 
```

And I can see the offset is 63 so I run:

```
┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/Xperience]
└─$ fls -o 63 xp_smol.vhd
d/d 3594-144-6: Documents and Settings
r/r 4-128-4:    $AttrDef
r/r 8-128-2:    $BadClus
r/r 8-128-1:    $BadClus:$Bad
r/r 6-128-1:    $Bitmap
r/r 7-128-1:    $Boot
d/d 11-144-4:   $Extend
```

Which is a full view of the file system. So now I just navigate using fls and extract using icat. First I go to the documents and settings located at 3594:

```
┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/Xperience]
└─$ fls -o 63 xp_smol.vhd 3594
d/d 3596-144-6: All Users
d/d 9843-144-5: daniel
d/d 3595-144-7: Default User
d/d 9866-144-6: LocalService
d/d 1680-144-6: NetworkService

┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/Xperience]
└─$ 
```

I see there is a user daniel so I navigate to that user with 9843:

```
┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/Xperience]
└─$ fls -o 63 xp_smol.vhd 9843
d/d 9984-144-1: Application Data
d/d 9983-144-1: Cookies
d/d 9982-144-1: Desktop
d/d 9981-144-6: Favorites
d/d 9966-144-6: Local Settings
d/d 9965-144-6: My Documents
d/d 9964-144-1: NetHood
r/r 9952-128-4: NTUSER.DAT
r/r 9953-128-4: NTUSER.DAT.LOG
r/r 10049-128-1:        ntuser.ini
d/d 9963-144-1: PrintHood
d/d 9962-144-5: Recent
d/d 9961-144-5: SendTo
d/d 9955-144-1: Start Menu
d/d 9954-144-6: Templates
r/r 10284-128-3:        wget.exe
r/r 10284-128-4:        wget.exe:Zone.Identifier

┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/Xperience]
└─$ 
```

Now I see a normal directory listing with folders like "My Documents." So I go there first:

```
┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/Xperience]
└─$ fls -o 63 xp_smol.vhd 9965
r/r 10056-128-1:        desktop.ini
d/d 10060-144-1:        My Music
d/d 10057-144-7:        My Pictures
d/d 10238-144-1:        My Videos
r/r 10229-128-5:        README.rtf

┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/Xperience]
└─$ 
```

And I find a README file which I extract using icat:

```
┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/Xperience]
└─$ icat -o 63 xp_smol.vhd 10229 > README

┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/Xperience]
└─$ cat README
{\rtf1\ansi\ansicpg1252\deff0\deflang1033{\fonttbl{\f0\fswiss\fcharset0 Arial;}}
{\*\generator Msftedit 5.41.15.1515;}\viewkind4\uc1\pard\f0\fs20 Welcome, CTF'ers to my Windows XP machine. I love XP because it's so simple, it provides an excellent place to make art and enjoy the wonderful digital environment which is 'bliss'. The only downside is that I was distracted when saving my masterpiece and now I don't remember where I saved it :(\par
}

┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/Xperience]
└─$ 
```

It gave me a solid hint to search for a "masterpiece" and knowing Windows XP it was probably a "mypaint" file. So I search the drive for the keyword "masterpiece" in the whole drive using the flag -r in the fls command:

```
┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/Xperience]
└─$ fls -r -o 63 xp_smol.vhd | grep -i "masterpiece"
+++++ r/r 10283-128-4:  masterpiece.bmp
+++ r/r 10289-128-1:    masterpiece.bmp.lnk
```

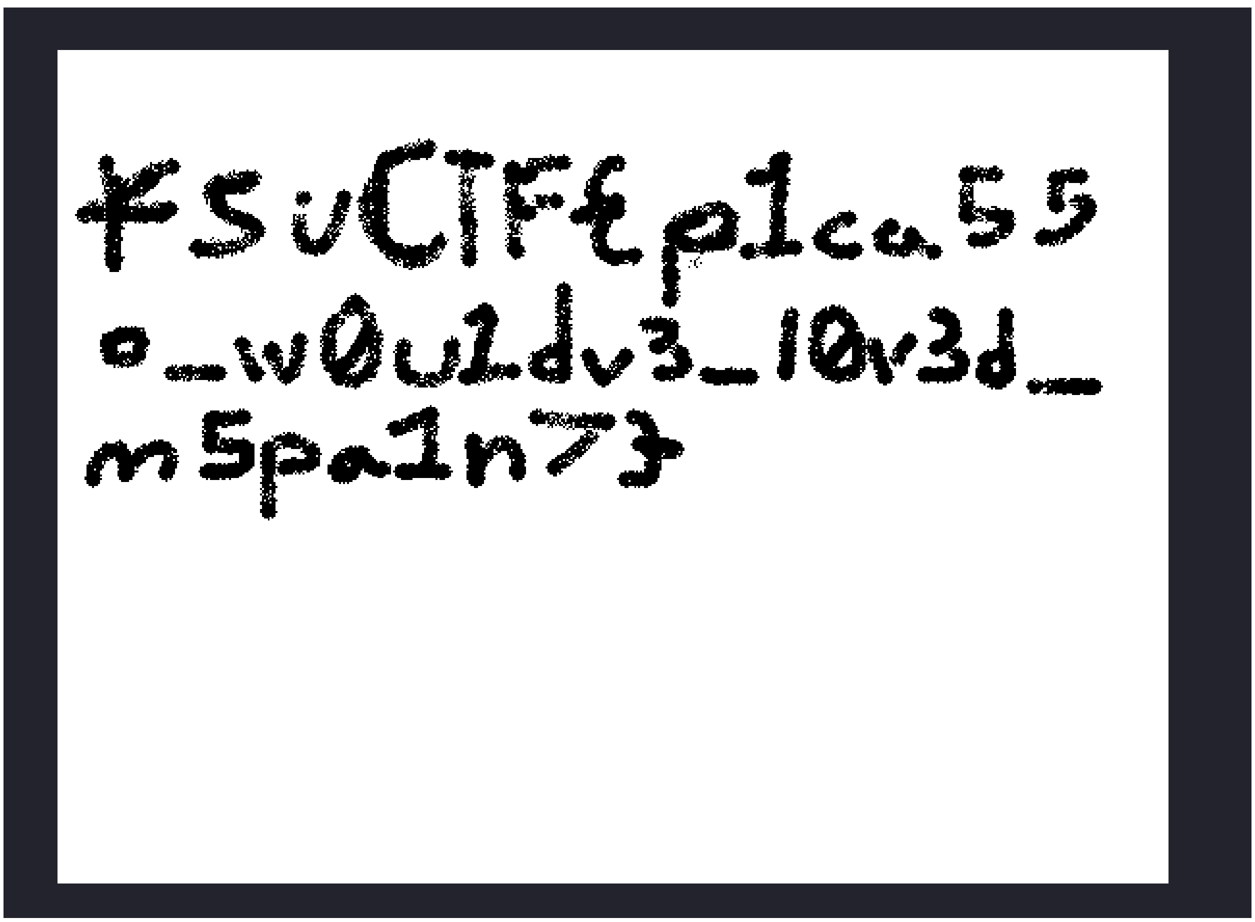
And it clearly shows a file titled masterpiece at location 10283 so I extract it using icat:

```
┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/Xperience]
└─$ icat -o 63 xp_smol.vhd 10283 > master.bmp

┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/Xperience]
└─$ ls
master.bmp  README  xp_smol.7z  xp_smol.vhd

┌──(kali㉿kali)-[~/CTFs/CTF_Class/forensics/Xperience]
└─$
```

And I open the picture:



And recover the flag: fsuCTF{p1ca55o_w0u1dv3_l0v3d_m5pa1n7}